COLIN NGUYEN

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ML researcher and AI engineer working on evaluation under uncertainty: how LLM agents and health-sensing models are measured, when a risk-control guarantee holds, and how to train models efficiently. Background in statistics.

RESEARCH EXPERIENCE

Evaluation of Mobile-Health Receptivity Models — with R. Benzo (OSU) and P. Washington (UCSF) *Jun 2026 – Present*

- Showed that pooled AUC, the standard metric for models predicting when someone will answer a health prompt, is dominated by between-person comparisons across three cohorts, so it mostly rewards knowing who tends to respond.
- Pre-registered a personalisation study (OSF) and trained a self-supervised GRU encoder on minute-level wearable data to test what its embeddings carry; results in the manuscript in preparation.

Risk-Controlled Decision Gates under Option-Count Shift — with Prof. Eric Suess, CSU East Bay *Oct 2026 – Present*

- Pre-registered study of whether a Learn-then-Test abstention guarantee (5% risk, 90% confidence) survives a change in option count: it fails from 5 to 150 options on a public decision model (7.2% risk) and, with its mechanism, on mine.
- Fine-tuned SmolLM2-135M/360M into decision models (anchored listwise loss, token-trie readout) matching the hosted model at every option count; four calibration regimes show per-count calibration underpowered and rank-normalisation leaky.

Efficient Training Methods for Language Models — with Prof. Eric Suess, CSU East Bay *Sep 2026 – Oct 2026*

- Five stacked memory-saving techniques (gradient checkpointing, 8-bit optimizer states, chunked loss, query-chunked attention, checkpointed loss chunks) raised SmolLM2's trainable step from 8,192 to 131,072 tokens in 71% of baseline memory, at a 17–22% throughput cost attributable entirely to checkpointing.
- Hand-wrote a block-wise 8-bit AdamW that matched torchao's optimizer state to the byte, and showed it equivalent to fp32 across six seeds at two model sizes under a pre-registered TOST margin.

The Ohio State University — Lifestyles Lab (PI: R. Benzo) *Jun 2026 – Present*

- Co-leading the analysis plan for validating consumer-wearable sleep measurement against research-grade actigraphy (night-matched Bland–Altman analysis, mixed models, TOST equivalence testing); also migrated the lab's data-collection tool from the retiring Fitbit API to the Google Health API.

Systematic Review: Computer Vision for Home-Video Phenotyping of ASD and ADHD — UCSF *Oct 2025 – Present*

- Screened studies for a ~55-study review (lead: Dr. Abdur Rasool; PI: Dr. Peter Washington); responsible for the meta-regression of model performance on video length, setting, validation and ground truth. *Manuscript in preparation.*

INDUSTRY EXPERIENCE

Athena Money — Founding AI Engineer *Nov 2025 – Present*

- Designed and built a multi-turn LLM evaluation platform for agents in regulated mortgage underwriting: scenario graders, rubric-based LLM-as-judge scoring, per-turn quality scoring and outcome checkpoints, driven by an LLM user simulator.
- Handled run-to-run variance with field-level consensus across repeated trials, confidence-score calibration and statistical regression tests gating tiered CI, with the evaluation driver on durable Temporal workflows for reproducible reruns.
- Built an experiments platform for comparing extraction and judgement approaches on regulated documents (LLM-as-judge scoring, field-level agreement across runs, QC datasets), which later fed the multi-turn evaluation framework.

Zeta Global — Data Science & GenAI Intern *Jun 2025 – Aug 2025*

- Multi-agent LLM orchestration (LangGraph) and an evaluation framework (accuracy / consistency / latency / cost).

NationBuilder — AI Research Intern *Jan 2025 – Nov 2025*

- RAG pipeline (MiniLM embeddings, FAISS) at 93% relevance accuracy; agentic document extraction for civic data.

SELECTED PROJECTS

NBA Catch Decisions — independent, on public player-tracking data *Oct 2026*

- Built a catch-level dataset from the only public NBA tracking data (621 games, 2015–16); showed players trade a foot of space at the catch for about eight seconds of shot clock, and that passed-up shots cannot be valued from shots taken.

SKILLS

Research methods: evaluation methodology, equivalence testing (TOST, Bland–Altman), risk control, calibration, pre-registration
ML / DL: PyTorch, fine-tuning / PEFT (LoRA, QLoRA), vLLM, scikit-learn, TensorFlow, Node2Vec, HuggingFace Transformers
LLM systems: RAG, LLM evaluation, LangGraph, embeddings, vector search (FAISS)
Languages / Tools: Python, R, SQL, Java, Go; AWS, Google Cloud, Docker, Spark, Databricks

EDUCATION

University of California, Santa Barbara *Sep 2022 – Jun 2025*
B.S. in Statistics & Data Science, Minor in Mathematics *Santa Barbara, CA*

PATENTS

- "Adaptive Neural Content Generation" (Zeta Global), U.S. Provisional App. No. 63/943,780, filed Dec 2025; first-named inventor.